POPULATION AND SYSTEMS ECOLOGY

# Module map

- Module 15: Population and Systems Ecology
- Connected lab: lab-16_population-systems-ecology.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 15
Population and Systems Ecology

1. Population growth models
2. Limits and carrying capacity
3. Community interactions
4. Energy flow through trophic levels
5. Matter cycling and systems feedback

**Lab evidence**
lab-16_population-systems-ecology.md

**Teaching note:** Open by naming the module claim and pointing students to the lab as the evidence surface.

POPULATION AND SYSTEMS ECOLOGY

# Learning objectives

- Compare exponential and logistic population growth.
- Explain carrying capacity and limiting factors.
- Distinguish density-dependent and density-independent regulation.
- Use trophic levels and the 10 percent rule to explain energy limits.
- Contrast energy flow with nutrient cycling.

**OBJECTIVE 1**
Compare exponential and logistic population growth.

**OBJECTIVE 2**
Explain carrying capacity and limiting factors.

**OBJECTIVE 3**
Distinguish density-dependent and density-independent regulation.

**OBJECTIVE 4**
Use trophic levels and the 10 percent rule to explain energy limits.

**OBJECTIVE 5**
Contrast energy flow with nutrient cycling.

**Teaching note:** Frame objectives as actions students should be able to perform, not facts to copy.

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# Topic sequence

- Population growth models: Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.
- Limits and carrying capacity: Exponential per-capita growth is constant; logistic per-capita growth slows near carrying capacity K. Total logistic growth peaks near K/2.
- Community interactions: Density-dependent effects vary with crowding. A freeze with equal proportional impact at low and high density is density-independent.
- Energy flow through trophic levels: Assume 10 percent transfer per trophic step for calculations; real efficiencies vary. From 10000 kJ in producers, primary consumers receive 1000 kJ.
- Matter cycling and systems feedback: Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

01

**Population growth models**

Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.

02

**Limits and carrying capacity**

Exponential per-capita growth is constant; logistic per-capita growth slows near carrying capacity K. Total logistic growth peaks near K/2.

03

**Community interactions**

Density-dependent effects vary with crowding. A freeze with equal proportional impact at low and high density is density-independent.

04

**Energy flow through trophic levels**

Assume 10 percent transfer per trophic step for calculations; real efficiencies vary. From 10000 kJ in producers, primary consumers receive 1000 kJ.

05

**Matter cycling and systems feedback**

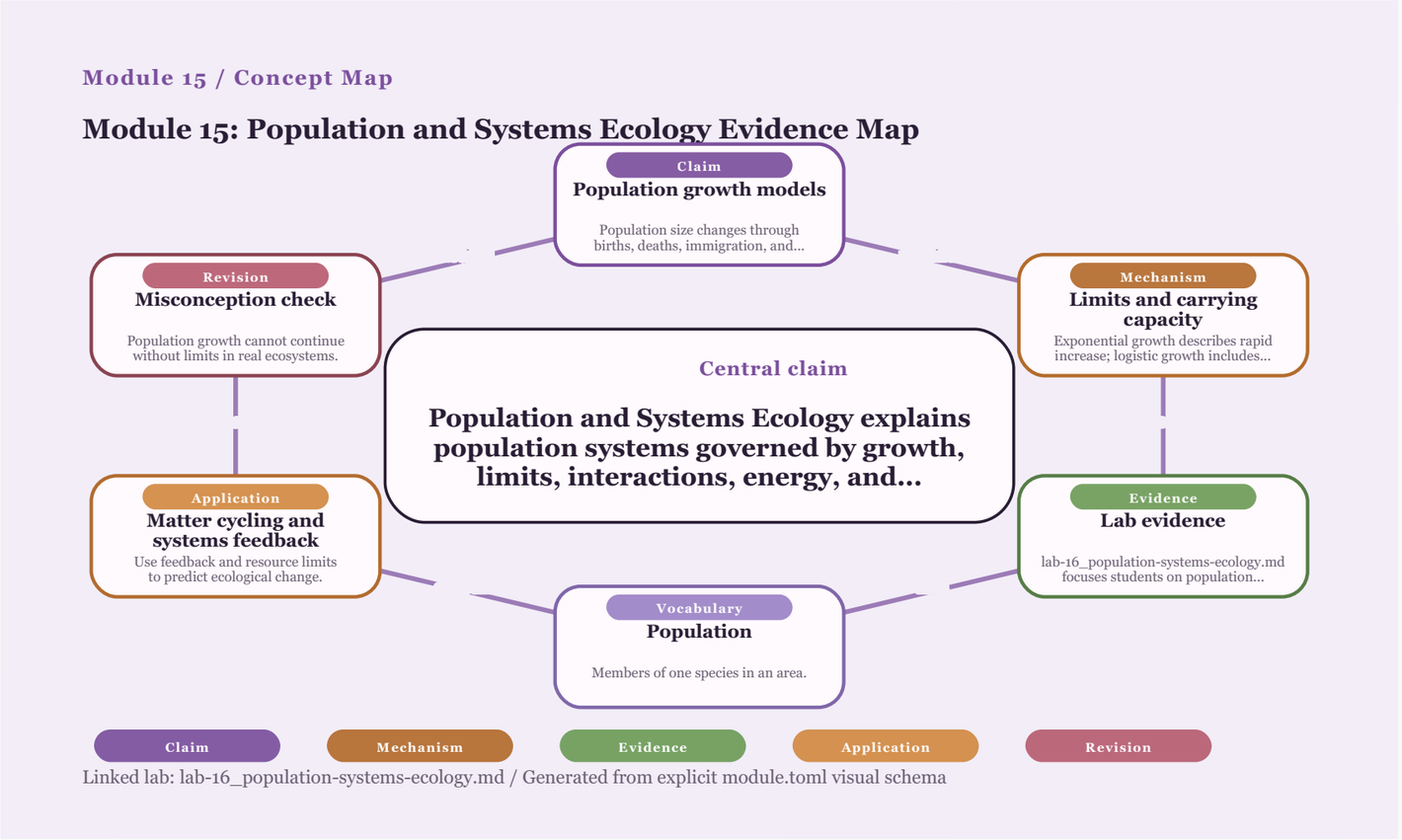
Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

**Teaching note:** Show the module as an ordered explanation so students can locate each new idea.

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# Module 15: Population and Systems Ecology Evidence Map

- Module focus: Population and Systems Ecology
- Central claim: Population and Systems Ecology explains population systems governed by growth, limits, interactions, energy, and feedback.
- Key nodes to connect: Population growth models, Limits and carrying capacity, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-16_population-systems-ecology.md supplies an evidence surface for the map.

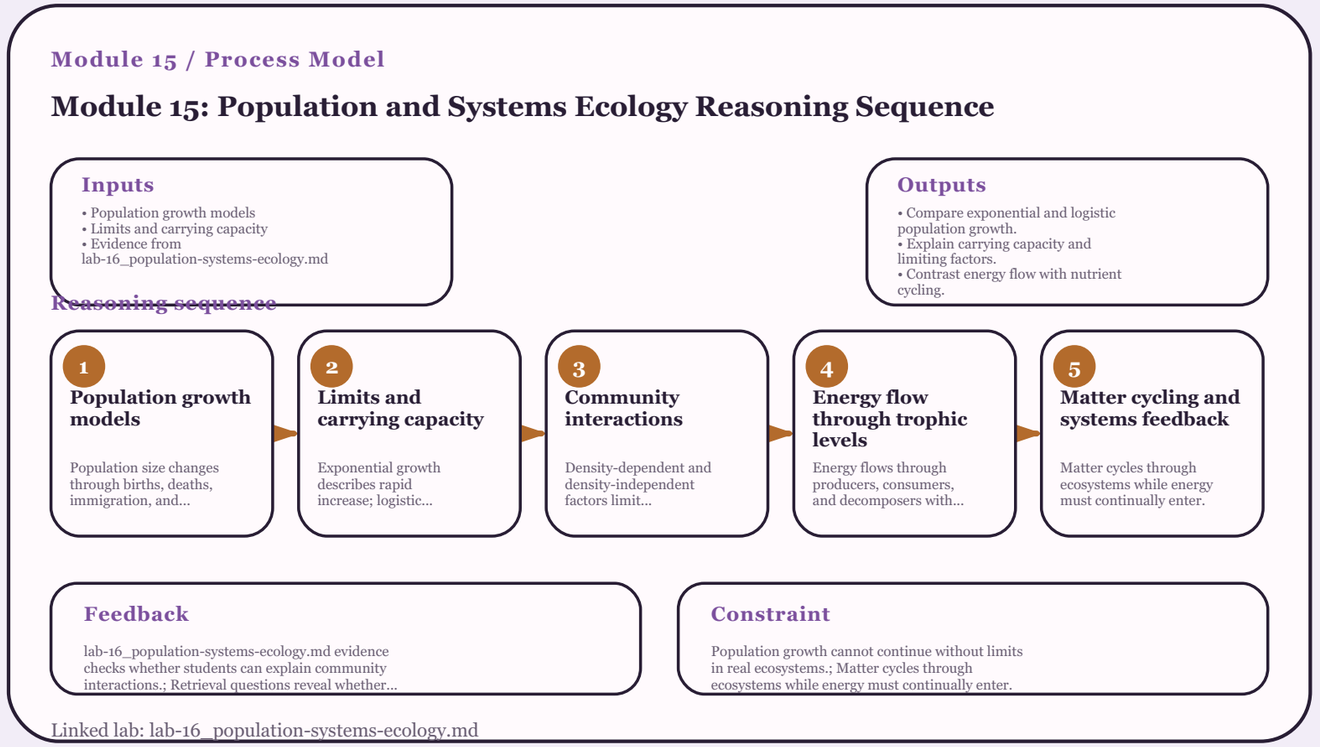



**Teaching note:** Ask students to explain one edge in their own words before moving on.

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# Module 15: Population and Systems Ecology Reasoning Sequence

- Module focus: Population and Systems Ecology
- Trace the model: Population growth models -> Limits and carrying capacity -> Community interactions -> Energy flow through trophic levels
- Inputs become outputs: Population growth models, Limits and carrying capacity -> Compare exponential and logistic population growth., Explain carrying capacity and limiting factors.
- Feedback check: lab-16_population-systems-ecology.md evidence checks whether students can explain community interactions.
- Lab connection: lab-16_population-systems-ecology.md gives students a process to observe or test.




**Teaching note:** Emphasize where the process can fail, slow down, or be revised by evidence.

POPULATION AND SYSTEMS ECOLOGY

# Terms as evidence handles

- Population: Members of one species in an area.
- Exponential growth: Growth pattern with increasing rate under abundant resources.
- Logistic growth: Growth that slows as population approaches carrying capacity.
- Carrying capacity: Approximate environment-supported population size.
- Density-dependent factor: Limiting factor whose impact changes with crowding.
- Density-independent factor: Limiting factor not directly tied to density.

**POPULATION**
Members of one species in an area.

**EXPONENTIAL GROWTH**
Growth pattern with increasing rate under abundant resources.

**LOGISTIC GROWTH**
Growth that slows as population approaches carrying capacity.

**CARRYING CAPACITY**
Approximate environment-supported population size.

**DENSITY-DEPENDENT FACTOR**
Limiting factor whose impact changes with crowding.

**DENSITY-INDEPENDENT FACTOR**
Limiting factor not directly tied to density.

**Teaching note:** Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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# Lab connection

- Lab file: lab-16_population-systems-ecology.md
- Lab evidence should test or illustrate: Community interactions
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**
Population growth models

**EVIDENCE**
lab-16_population-systems-ecology.md

**CLAIM**
Compare exponential and logistic population growth.

**Teaching note:** Make the lab connection explicit before students encounter the activity.

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# Contrast check

- Do not stop at: Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.
- Stronger explanation: Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.

**DEEPER**

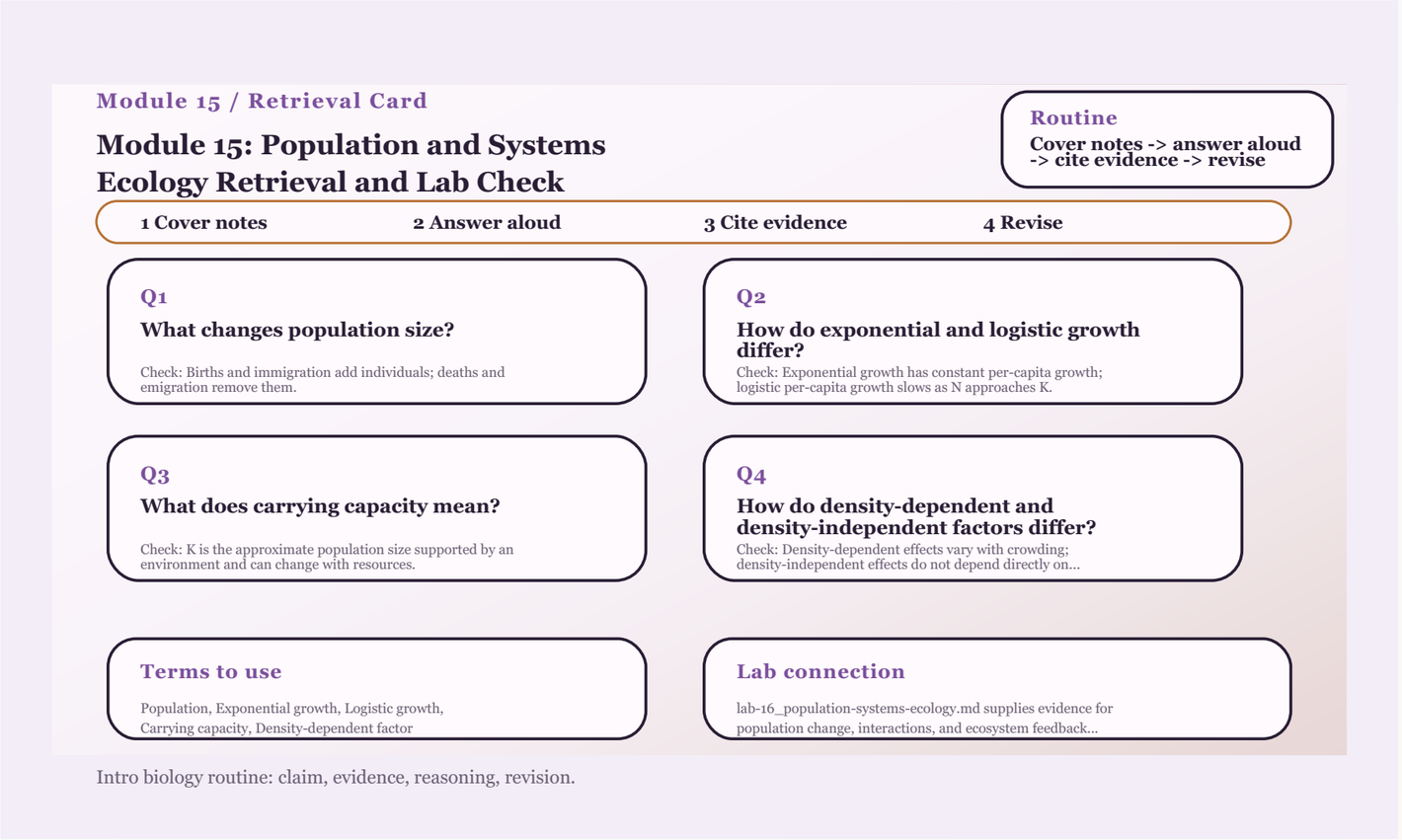
Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

**Teaching note:** Use this slide to prevent shallow vocabulary-only learning.

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# Module 15: Population and Systems Ecology Retrieval and Lab Check

- Module focus: Population and Systems Ecology
- Retrieval prompt: What changes population size?
- Answer check: Births and immigration add individuals; deaths and emigration remove them.
- Required terms: Population, Exponential growth, Logistic growth
- Lab connection: lab-16_population-systems-ecology.md; lab-16_population-systems-ecology.md supplies evidence for population change, interactions, and ecosystem feedback data.




Intro biology routine: claim, evidence, reasoning, revision.

**Teaching note:** Pause here. Students answer first without notes, then compare to the checks.

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# Practice quiz bridge

- 1. Logistic growth differs from exponential growth because logistic growth:
- 2. A contagious disease that spreads faster when a population is crowded is an example of a:
- 3. About how much of the energy at one trophic level typically passes to the next level up?
- 4. Energy and matter behave differently in an ecosystem because:

**QUESTION 1**

**Key: C**

Logistic growth differs from exponential growth because logistic growth:

**QUESTION 2**

**Key: A**

A contagious disease that spreads faster when a population is crowded is an example of a:

**QUESTION 3**

**Key: D**

About how much of the energy at one trophic level typically passes to the next level up?

**QUESTION 4**

**Key: B**

Energy and matter behave differently in an ecosystem because:

**Teaching note:** Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

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# Synthesis and exit ticket

- Exit claim: explain population growth models using evidence from lab-16_population-systems-ecology.md.
- Must include: Population, Exponential growth, Logistic growth.
- Revision prompt: what evidence would change your explanation?

**CLAIM**
Population growth models

**EVIDENCE**
lab-16_population-systems-ecology.md

**REVISION**
What would change your mind?

**Teaching note:** End with a claim-evidence-reasoning exit ticket connected to the same module data.